Machine Learning Course Notes

Train/test split

We always hold back part of the data as a test set. The model never sees it during training, so its score on the test set is an honest estimate of how it will do on new data. A typical split is 75 percent for training and 25 percent for testing.

Overfitting

Overfitting is when a model memorises the training rows instead of learning the pattern. The sign is a large gap between train accuracy and test accuracy, for example train 1.000 but test 0.771. A model that is perfect on training data but weak on new data has memorised, not learned.

Cross-validation

Cross-validation splits the data into k folds, trains on k minus one of them and tests on the held-out fold, then rotates and averages. It gives a steadier, more honest score than a single split, reported as mean plus or minus standard deviation.

Data leakage

Leakage happens when information from the test set sneaks into training, for example fitting a scaler on all the data before splitting. The fix is to put every transformer inside a Pipeline so it is fitted on the training fold only.

Decision tree

A decision tree asks a series of yes or no questions about one feature at a time, such as is worst radius less than seventeen. It chooses each question by measuring Gini impurity and keeping the split that makes the child groups purest. Its strength is that you can read it.

Gini impurity

Gini impurity measures how mixed a group is. The formula is one minus the sum of squared class fractions. Zero means the group is pure, all one class, while zero point five means an even fifty fifty mix. A tree picks the split that most reduces impurity.

Random forest

A random forest grows many decision trees, each on a random sample of rows and features, then lets them vote. Because the trees make different mistakes, the errors cancel out and the forest generalises better than any single tree. This trick is called bagging.

Feature importance

Feature importance ranks how much each column contributed to a tree model's decisions, by totalling the impurity reduction from every split that used it. It turns a model into a story you can tell, for example distance and preparation time drove most predictions.

k-means clustering

k-means is unsupervised, meaning it has no labels. It places k centres at random, assigns every point to its nearest centre, moves each centre to the average of its points, and repeats until nothing moves. It minimises inertia, the total squared distance to centres.

Choosing k

To choose the number of clusters, use the elbow method, where you plot inertia against k and look for the bend, and the silhouette score, which rates how cleanly separated the clusters are from minus one to plus one. When they disagree, you decide using domain knowledge.

Feature engineering

Feature engineering means creating better input columns, such as ratios, differences or date parts. A model can only use what you give it. Adding one good column can help far more than switching to a fancier model.

Scaling

Standard scaling rewrites each value as z equals x minus the mean divided by the standard deviation. Distance-based models such as logistic regression, k-means and neural networks need it, because otherwise a large-valued column dominates. Trees and forests do not need it.

One-hot encoding

One-hot encoding turns a text column into one zero-or-one column per category. You must not simply number the categories one, two, three, because that invents a false ordering that the model would wrongly trust.

The neuron

A neuron does two steps. First a weighted sum of its inputs plus a bias, and then an activation function that bends the result. A single neuron with a sigmoid activation is exactly logistic regression.

Activation functions

An activation adds a bend so a network can do more than draw a straight line. Sigmoid squashes any number into zero to one and suits a final probability. ReLU is max of zero and x, and is the usual choice inside hidden layers.

Neural network layers

A network stacks neurons into layers: an input layer, one or more hidden layers, and an output layer. Hidden layers build their own features from the raw inputs, which is why a network can learn a curved boundary that a straight line cannot.

Training a network

Training repeats four steps: a forward pass to get a prediction, a loss to measure how wrong it is, backpropagation to find which weights caused the error, and gradient descent to nudge each weight a small step downhill. Repeat thousands of times.

Learning rate

The learning rate is the step size in gradient descent, written new weight equals old weight minus rate times slope. Too small and training crawls and never arrives. Too big and it overshoots and the loss bounces instead of falling.

Early stopping

Early stopping holds out a small validation slice during training and halts when that score stops improving. It prevents the network from carrying on into memorisation, shrinking the train-test gap and saving a lot of training time.

Bag of words

Bag of words turns text into numbers by giving each vocabulary word a column and counting occurrences. It works surprisingly well, but it throws away word order, so the phrase not good looks almost identical to good.

TF-IDF

TF-IDF weights each word by how often it appears in a document times how rare it is across all documents. Common words such as the get crushed toward zero because they carry little meaning, while distinctive words get boosted.

Embeddings

An embedding represents a word or document as a short list of numbers, positioned so that similar meanings sit close together. Unlike bag of words, it captures that good and great are related. Relationships become arithmetic, for example king minus man plus woman equals queen.

Cosine similarity

Cosine similarity compares the direction of two vectors, computed as their dot product divided by both their lengths. One means the same meaning, zero means unrelated. We use direction rather than distance so document length does not matter.

Semantic search

Semantic search embeds every document once, embeds the incoming query the same way, scores each document by cosine similarity, and returns the highest scoring ones. It finds matches by meaning rather than by exact keyword.